COMSATS University Islamabad, Vehari Campus

Department of Computer Science

Lab Mid

Fall-2026

CSC336 Web Technologies

Class: BSSE **Instructor: Yasmeen Jana** **Lab Mid given: 25/10/2026**

Marks: 30 **Submission Date: 4/11/2026**

Name: _____ **Reg. No: _____**

CLO NO	Course Learning Outcome	Blooms Taxonomy Learning Level
CLO4	Apply the concepts of markup & scripting languages and client-side technologies.	Applying

Note: Submit all source files and a complete demo video demonstrating the implemented features and a full game session for each mode.

Build a responsive Typing Challenge Game using HTML5, CSS3, and JavaScript (ES6+). You may use Bootstrap or Tailwind CSS for styling. The game must provide distinct gameplay rules, controls, scoring, feedback, and completion conditions for every implemented mode.

1. Required Game Modes

Mode A: Bubbles (Letter Recognition)

- Display at least five bubbles simultaneously. Each bubble must contain a letter, and bubbles must move upward at a controlled speed.
- When the player presses a key matching a visible bubble, remove the matching bubble and increase the score.
- Provide immediate visual feedback for correct and incorrect keypresses. Define and apply a penalty for an incorrect keypress.
- Count a miss when a bubble reaches the top or another clearly defined boundary. End the game when the configured miss limit is reached (for example, six misses).
- Increase difficulty during play by raising bubble speed, increasing the number of active bubbles, or applying another clearly defined difficulty change.

Mode B: KeyMagnet (Keyboard Precision)

- Display a target key and a visible keyboard layout or simplified key panel.
- Generate target keys from a JavaScript array. Include at least one practice set, such as the home-row keys A, S, D, F, J, K, and L.
- Award points for correct responses and show immediate feedback when an incorrect key is pressed.
- Track correct responses, incorrect responses, and response time.

- Allow the player to select a key set or difficulty level, and ensure the selection changes actual gameplay.

2. Mandatory Functional Requirements

#	Requirement	Specification
1	Game selection	Provide a home screen for each implemented mode, including its name, a short description, and a Start button.
2	Difficulty levels	Provide Easy, Medium, and Hard. Each level must change gameplay in a meaningful way, such as speed, target complexity, sequence length, or time limit.
3	Gameplay controls	Provide working Start, Restart, and Reset controls. These controls must operate without reloading the page.
4	Score and status	Display the current score and relevant status information, such as remaining time, lives, misses, difficulty, or correct and incorrect keypresses.
5	Immediate feedback	Visually distinguish correct and incorrect actions and provide feedback promptly.
6	Game completion	Define a clear win, completion, timeout, or game-over condition for every mode.
7	Results screen	At the end of each session, display the score and relevant performance statistics for that mode.
8	State reset	Restarting must reset all applicable state, including score, timer, lives, targets, input, and mode-specific statistics.
9	Timer and movement management	Stop timers and repeated movement when a game ends or restarts. Prevent multiple timers or animation loops from running simultaneously.
10	Responsive interface	Keep the layout and controls usable on desktop and mobile-sized screens.

3. Assessment and Marks Distribution

Assessment Component	What will be assessed	Marks
Game modes and functionality	Both modes work as specified; controls, scoring, feedback, difficulty, and end conditions function correctly.	10
Mandatory requirements and state management	Results screen, reliable restart/reset, timer cleanup, status indicators, and no page reloads or duplicate loops.	5
UI/UX, responsiveness, and originality	Clear, polished, original interface that remains usable on desktop and mobile.	5
Demo video	A complete video demonstration showing the home screen, both modes, difficulty selection, gameplay, feedback, controls, and end-of-game results.	5
Viva	Understanding and modification in code	5
Total		**30**